

Recall from last class:

- A set of vectors  $\{\vec{v}_1, \dots, \vec{v}_p\}$  in  $\mathbb{R}^n$  is said to be **linearly independent** if

$$x_1\vec{v}_1 + x_2\vec{v}_2 + \dots + x_p\vec{v}_p = \vec{0}$$

has only the trivial solution. The set is called **linearly dependent** if the equation has non-trivial solutions and the equation is called the **dependence relation**.

- Let  $A = [\vec{a}_1 \dots \vec{a}_n]$ , then  $\{\vec{a}_1 \dots \vec{a}_n\}$  are linearly independent if and only if  $A\vec{x} = \vec{0}$  has only the trivial solution.
- **Theorem 7:** A set  $S = \{\vec{v}_1, \dots, \vec{v}_p\}$  of two or more vectors is linearly dependent if and only if one of the vectors of  $S$  is a linearly combination of the others.

Moreover if  $\vec{v}_1 \neq \vec{0}$ , then some  $\vec{v}_j$  for  $j > 1$  is a linear combination of  $\vec{v}_1, \dots, \vec{v}_{j-1}$ .

- **Uri's Theorem:** Any set  $\{\vec{v}_1, \dots, \vec{v}_p\}$  in  $\mathbb{R}^n$  is linearly dependent if  $p > n$ .

Define  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  by  $T(\vec{x}) = A\vec{x}$  where  $A$  is an  $m$  by  $n$  matrix. Recall we call  $\mathbb{R}^n$  the domain and  $\mathbb{R}^m$  the codomain. The set of outputs,  $\{T(\vec{x}) : \vec{x} \in \mathbb{R}^n\}$ , is the image or the range of  $T$ .

**Definition 1.**  $T$  is **linear** if

1.  $T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$  for all  $\vec{u}, \vec{v}$  in  $\mathbb{R}^n$ ,
2.  $T(c\vec{u}) = cT(\vec{u})$  for all  $\vec{u}$  in  $\mathbb{R}^n$  and  $c$  in  $\mathbb{R}$ .

### 1. Properties of Linear Transformations.

(a) Prove if  $T$  is linear, then  $T(\vec{0}) = \vec{0}$ .

(b)  $T(c\vec{u} + d\vec{v}) = cT(\vec{u}) + dT(\vec{v})$ .

2. Let  $T(\vec{x}) = A\vec{x}$  where  $A = \begin{bmatrix} 1 & 0 \\ -3 & 1 \\ 2 & -2 \end{bmatrix}$  and let  $\vec{b} = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$ .

(a) Is  $\vec{b}$  in the range of  $T$ ?

(b) What is the range of  $T$ ?

(c) Is  $T$  surjective?

(d) Is the transformation  $T$  injective?

3. Let  $T(\vec{x}) = A\vec{x}$  where  $A = \begin{bmatrix} 1 & 0 & -3 \\ -3 & 1 & 6 \\ 2 & -2 & -1 \end{bmatrix}$  and let  $\vec{b} = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$ . Is  $\vec{b}$  in the range of  $T$ ?

4. Let  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  be linear and  $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$  be a linearly dependent set. Is  $\{T(\vec{v}_1), T(\vec{v}_2), T(\vec{v}_3)\}$  linearly dependent?

# Linear Transformations – Answers / Solutions

1.

- (a) Suppose  $T$  is linear, then

$$T(\vec{0}) = T(0 \cdot \vec{u}) = 0 \cdot T(\vec{u}) = \vec{0}.$$

- (b) Suppose  $T$  is linear. By property (1) of linearity, we know

$$T(c\vec{u} + d\vec{v}) = T(c\vec{u}) + T(d\vec{v}).$$

By property (2), we know

$$T(c\vec{u}) + T(d\vec{v}) = cT(\vec{u}) + dT(\vec{v}).$$

Thus we have the result.

2.

- (a) In order to determine if  $\vec{b}$  is in the range of  $T$ , we solve the matrix equation  $A\vec{x} = \vec{b}$ . Forming the augmented matrix and row-reducing, we get

$$\left( \begin{array}{cc|c} 1 & 0 & -2 \\ -3 & 1 & 3 \\ 2 & -2 & -1 \end{array} \right) \sim \left( \begin{array}{cc|c} 1 & 0 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & -3 \end{array} \right).$$

Thus there is no solution to  $A\vec{x} = \vec{b}$ .

- (b) Replacing  $\vec{b}$  for the previous part with arbitrary components will give the range of  $T$ . We get

$$\left( \begin{array}{cc|c} 1 & 0 & b_1 \\ -3 & 1 & b_2 \\ 2 & -2 & b_3 \end{array} \right) \sim \left( \begin{array}{cc|c} 1 & 0 & b_1 \\ 0 & 1 & b_2 + 3b_1 \\ 0 & 0 & b_3 + 2b_2 + 4b_1 \end{array} \right).$$

Thus  $\vec{b}$  is in the range of  $T$  if and only if  $b_3 + 2b_2 + 4b_1 = 0$ . We can see that the range is a plane in  $\mathbb{R}^3$ .

- (c) No,  $T$  is not surjective. In particular, we showed in part (a) that there is no element in the domain that will map to  $\vec{b} = \begin{bmatrix} -2 \\ 3 \\ -1 \end{bmatrix}$  under  $T$ . Equivalently we can say that  $T(\vec{x}) = \vec{b}$  has no solutions.
- (d) Recall  $T$  is injective if for all  $\vec{b}$ , there is at most one  $\vec{x}$  such that  $T(\vec{x}) = \vec{b}$ . We use the contrapositive form of injectivity. Suppose  $T(\vec{x}) = T(\vec{y})$ , then we get  $A\vec{x} = A\vec{y}$ . Rearranging the equation we get  $A\vec{x} - A\vec{y} = \vec{0}$ , and factoring, we get  $A(\vec{x} - \vec{y}) = \vec{0}$ . We want to show that  $A\vec{z} = \vec{0}$  has only the trivial solution. Forming the augmented matrix and row-reducing, we find  $A\vec{z} = \vec{0}$  has only the trivial solution. Therefore  $A(\vec{x} - \vec{y}) = \vec{0}$  if and only if  $\vec{x} - \vec{y} = \vec{0}$ , and hence  $\vec{x} = \vec{y}$ . Thus  $T$  is injective.

3. Yes,  $\vec{b}$  is in the range of  $T$ . In order to determine if  $\vec{b}$  is in the range of  $T$ , we solve the matrix equation  $A\vec{x} = \vec{b}$ . Forming the augmented matrix and row-reducing, we get

$$\left( \begin{array}{ccc|c} 1 & 0 & -3 & -2 \\ -3 & 1 & 6 & 3 \\ 2 & -2 & -1 & -1 \end{array} \right) \sim \left( \begin{array}{ccc|c} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & 3 \end{array} \right).$$

Thus there is a unique solution to  $A\vec{x} = \vec{b}$ .

4. Let  $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$  be linear and  $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$  be a linearly dependent set. Since  $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$  is linearly dependent, we know there exists  $c_1, c_2, c_3$ , not all zero, such that  $c_1\vec{v}_1 + c_2\vec{v}_2 + c_3\vec{v}_3 = \vec{0}$ . Applying  $T$  to both sides of the equation we get

$$T(c_1\vec{v}_1 + c_2\vec{v}_2 + c_3\vec{v}_3) = T(\vec{0}).$$

Using linearity of  $T$ , we get

$$c_1T(\vec{v}_1) + c_2T(\vec{v}_2) + c_3T(\vec{v}_3) = \vec{0}.$$

Since not all  $c_1, c_2, c_3$  are zero, we have a non-trivial solution to the vector equation

$$x_1T(\vec{v}_1) + x_2T(\vec{v}_2) + x_3T(\vec{v}_3) = \vec{0},$$

and hence  $\{T(\vec{v}_1), T(\vec{v}_2), T(\vec{v}_3)\}$  is linearly dependent.